

Cirsium pannonicum
purple / upright; YOLO p=0.88



Cirsium mohavense
pale / inclined; YOLO p=0.98



Cirsium subcoriaceum
pink / inclined; YOLO p=0.85

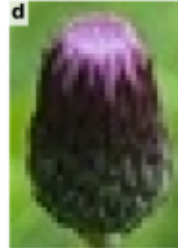


Actual production records

- source photos used in the executed atlas
- stored YOLO visible-capitulum boxes
- open photo licences verified
- final v2 continuous measurements

YOLO defines the observation unit.
Colour, outline and orientation are then measured by deterministic image processing.

Tight head crop



Context crop



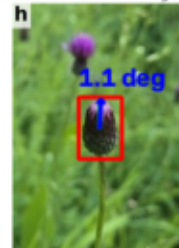
Corolla colour mask



Outline + convex hull



Orientation relative to image vertical



i Continuous outputs for the detailed example

Orientation	0.7°	status: usable
Lightness	56.5	status: usable
Chroma	33.3	status: usable
Hue	300.2°	status: usable
Aspect ratio	1.59	status: usable
Circularity	0.77	status: usable
Solidity	0.95	status: usable

Detailed example: *Cirsium pannonicum*; observation 295734197; photo 532420148; (c) Bobo-X, some rights reserved (CC BY); licence CC-BY

27 registered

continuous endpoints

22 measured

finite v2 outputs

5 unexecuted

remain missing evidence

Localization only

detector does not score traits

Three open-licensed photographs are presentation/source provenance only; the v2 cohort and results come exclusively from the frozen full-27 lane.